

Introduction to Microeconomics

Lecture 13—Price Indexes

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cost-of-living index or price index Ratio of the present cost of a typical bundle of consumer goods and services compared with the cost during a base period.

ideal cost-of-living price index Cost of attaining a given level of utility at current prices relative to the cost of attaining the same utility at base-year prices.

- The ideal cost-of-living price index takes into consideration that consumers **substitute** towards relatively cheaper consumption bundles of goods and services when facing different price ratios across periods.

Laspeyres and Paasche Indexes

For a set of goods and services $\mathcal{C} = \{c_1, c_2, \dots, c_k\}$, let

- $p_{c,t}$: prevailing price of good c in period t .
- $q_{c,t}$: quantity of c consumed in period t .

Let t_n denote the current time, and t_0 denote the **base year**.

- Laspeyres Index

$$\mathcal{L}(t_0, t_n, q_{c,t_0}) = \frac{p_{c_1,t_n} q_{c_1,t_0} + p_{c_2,t_n} q_{c_2,t_0} + \dots + p_{c_k,t_n} q_{c_k,t_0}}{p_{c_1,t_0} q_{c_1,t_0} + p_{c_2,t_0} q_{c_2,t_0} + \dots + p_{c_k,t_0} q_{c_k,t_0}}$$

- Paasche Index

$$\mathcal{P}(t_0, t_n, q_{c,t_n}) = \frac{p_{c_1,t_n} q_{c_1,t_n} + p_{c_2,t_n} q_{c_2,t_n} + \dots + p_{c_k,t_n} q_{c_k,t_n}}{p_{c_1,t_0} q_{c_1,t_n} + p_{c_2,t_0} q_{c_2,t_n} + \dots + p_{c_k,t_0} q_{c_k,t_n}}$$

- In general, a price-index depends on the base year t_0 , the current year t_n , and the basket of goods q_c .

The ideal cost of living index: example

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

The ideal cost of living index: using the expenditure minimization problem to find the bundle in the new year

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

The ideal cost of living index: Computing the index

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

Computing the Laspeyres Index

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

Computing the Paasche Index

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

- We have that in general:

$$\mathcal{P} \leq \mathcal{I} \leq \mathcal{L}$$

- Potential remedy is to compute **chain-weighted indexes**:
 - For the Laspeyres index:

$$\mathcal{L}(t_0, t_1, q_{c,t_0}) \times \mathcal{L}(t_1, t_2, q_{c,t_1}) \times \cdots \times \mathcal{L}(t_{n-1}, t_n, q_{c,t_{n-1}})$$

- For the Paasche index:

$$\mathcal{P}(t_0, t_1, q_{c,t_1}) \times \mathcal{P}(t_1, t_2, q_{c,t_2}) \times \cdots \times \mathcal{P}(t_{n-1}, t_n, q_{c,t_n})$$

- An example of the Laspeyres index is the CPI (Consumer Price index): Fix a base year basket and a base year and track its index across time. The CPI is used to measure inflation in all countries around the world.
- Example of a Paasche index: GDP Deflator, which is the ratio of Nominal GDP/Real GDP. As you can see this index fixes a base year and a current year basket.
 - Nominal GDP: value of goods and services at current prices.
 - Real GDP: value of goods and services at base year prices.